NATURAL LANGUAGE PROCESSING I
DEEP LEARNING

ADVERSARIAL ATTACKS ON NLP MODELS
PROJECT PROPOSAL

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Introduction

Deep Neural Networks (DNNs) are large Neural Networks (NNs) whose architecture is organised as a series of layers of neurons. They are widely used nowadays, as they can propose a solution to complicated problems that cannot be modelled as linear or non-linear problems. Although extremely useful, they carry a fundamental problem: interpretability. These techniques currently work as a black box. It is difficult to determine why a DNN has made a particular prediction.

Interpretability is compulsive at both extremes of performance. This area was firstly developed due to the population's need to comprehend why a certain model had failed given a certain input. For example, an important part of why these models are not widely implemented in the sanitary field, although they show above-human performance, is the lack of reasons one can give for a failed categorisation. On the other hand, the performance of the state-of-the-art models is often inexplicably high. Nevertheless, even those models are vulnerable to attacks that can lead them to underperform. These attacks may be unnoticeable for the human eye, but DNNs are very susceptible to them.

These reasons set the ground for the study of Adversarial Attacks, as well as possible defences from them. Ultimately, it is an area that studies the robustness of DNNs.

In this proposal we will briefly mention different text attacks and defence tactics.

Text Attacks and Possible Defences

Over the past few years, numerous methods for adversarial attacks have been introduced, which are specifically designed for NLP tasks as computer vision examples are fundamentally different.

The aim of the adversarial attacks is to manipulate inputs of a multi-class classifier in such a manner that they are misclassified with a high degree of confidence. Considering a classification system with k distinct classes denoted by $C_1, C_2, \dots, C_k$ an adversarial example x' is crafted from the original input x. Let f be the discriminant function demarcating the decision boundaries for the classifier. If x is an instance for class C_i and the adversarial target is C_{target}, the attack's objective is to achieve:

$$f_{C_{target}}(x') > f_i(x')$$

Under this formulation, adversarial attacks can be framed as an optimization problem for the adversarial example x':

$$\min_{x'} |\, x' - x|$$

Subject to the restriction:

$$\max_{i \neq \text{target}} f_i\,(x') - f_{\text{target}}(x') \leq 0$$

The inequality constraint illustrates the core of an adversarial attack – to modify the input x in a way that the classifier prefers the target class over the true class. The adversarial problem aims to minimize the perturbation size, increasing the stealth of the attack while ensuring that the modified input x' is identified as belonging to C_{target}.

These attacks involve various strategies designed to manipulate NLP models. There are different granularity levels depending on what the attack is focused on: **character-level** (a character is changed), **word-level** (a word is changed, for example, using a synonym) and **sentence-level** (for example changing the order of the words in a sentence). **Semantic attacks** aim to change the meaning of the text while keeping it grammatically correct. On the other hand, **syntactic attacks** focus on altering the structure or grammar while preserving its meaning. Lastly, **char-level manipulations** include character swapping, repeating, substitution, deletion and insertion. **Obfuscation attacks** introduce noise or distortions combining semantic, syntactic and char-level attacks to prevent accurate analysis, whilst **evasion attacks** try to avoid the model's defences without changing the meaning. **Poisoning attacks** manipulate the model behaviour by injecting malicious data while training the model. These attacks can use **gradient-based methods**, where attackers use gradient information to generate adversarial examples; **or black-box strategies**, where attackers rely on heuristics to generate adversarial examples. The attacker's level of knowledge, from zero-knowledge to full knowledge, further categorizes these attacks, each presenting different challenges and implications for NLP security.

To address the growing problem of adversarial attacks on natural language processing (NLP) systems, researchers have developed various defence mechanisms. One prominent approach is adversarial training, where models are trained on crafted examples to improve their resilience. Other techniques like robust preprocessing and ensemble methods also help mitigate the impact of adversarial inputs. Methods like feature squeezing and defensive distillation manipulate input features or training data to make models less vulnerable to small changes. Certified defences provide formal guarantees on model robustness, while randomization introduces unpredictability in model predictions. Adversarial training combined with transfer learning uses knowledge from related domains to enhance generalization. Input transformation techniques alter representations to frustrate adversarial attacks. Finally, robust optimization algorithms directly optimize models for robustness during training, collectively forming a comprehensive arsenal of defenses against NLP adversarial threats.

Milestones

12th of April

Finished code. Implement different attacks and defences. Check how different models react to our attacks. Try to make them more robust to the vulnerabilities shown

19th of April

Finished paper. Prepare presentation.

Project Plan

For the attacks, we would like to use different type of model and for this we will use our own models and pretrained ones.

There are many ways to attack or defend a model. We would like to try different attacks such as semantic, syntactic or poisoning and analyse how each attack works and its effect on the model. On the other hand, we would like to find how to make models robust against each type of attack, with the methods mentioned before like adversarial training.

The number of attacks programmed in this project will depend on how deep we go into each one.

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